

Mikel Sagardia, PhD

Principal Data Scientist and Machine Learning Engineer

I have 15+ years of experience building and scaling production systems in computer vision, multimodal AI, and robotics, with a proven record of turning research into deployed, business-relevant AI products across telecom, manufacturing, and aerospace.

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Work Experience

Principal Data Scientist and ML Engineer @ Deutsche Telekom Bonn (remote from San Sebastian), 2023.05 — present

- Owned end-to-end design and delivery of production-grade ML libraries for computer vision, text, and 3D data, supporting multimodal processing, segmentation, detection, vectorization, clustering, and search.
- Architected scalable multimodal ML applications that automated key business processes, delivering ~30% efficiency gains.
- Drove enterprise adoption of generative AI by identifying high-impact use cases and translating them into deployable solutions.
- Designed, benchmarked, and deployed LLM, RAG, and agent-based applications with a strong focus on evaluation, robustness, and production readiness.
- Onboarded and mentored team members, accelerating ramp-up and establishing shared best practices for production ML and AI.

Computer Vision and Machine Learning Engineer @ Self Employed San Sebastian, 2022.04 — 2023.04 (1 year, 1 month)

- Delivered production-grade computer vision and machine learning solutions for industrial clients as an independent consultant.
- Advised and trained engineering teams on deep learning best practices, model development, and deployment workflows.

Senior Researcher @ Vicomtech San Sebastian, 2020.05 — 2021.12 (1 year, 8 months)

- Led and delivered European and national research projects in computer vision, machine learning, and robotics.
- Drove acquisition of new funded projects by defining technical proposals and research direction.
- Supervised and mentored MSc and PhD students, with theses resulting in peer-reviewed publications.

Project Engineer in Computer Vision @ MVTec Munich, 2019.03 — 2020.03 (1 year, 1 month)

- Acted as technical lead for customer-facing machine vision projects, bridging product capabilities and industrial requirements.
- Trained customers and global distributors on 3D vision and robotics applications, acting as technical subject-matter expert.
- Presented new HALCON features at MVTec Innovation Day 2020, supporting product adoption and customer engagement.
- Provided advanced technical support across a broad range of industrial machine vision use cases.

Research Engineer @ German Aerospace Center (DLR) Weßling / Munich, 2008.04 — 2018.12 (10 years, 9 months)

- Led a 5-year project across two distributed teams, developing a virtual reality framework for satellite on-orbit servicing which combined robotics systems, haptic rendering, and real-time simulation methods.
- Researched and implemented collision detection and force computation methods for real-time haptic rendering.
- Coordinated industry collaboration projects with partners (VW, Bosch), translating research prototypes into applied systems.
- Published 22+ peer-reviewed papers in top-tier venues (IEEE, ACM), contributing foundational work in haptics and robotics.
- Mentored and supervised 20+ MSc students and interns, supporting technical development and research quality.

Education

PhD in Electrical and Information Engineering @ Technische Universität München, TUM Munich, 2013.12 — 2018.11

Thesis (completed while at DLR; defended in 2019): *Virtual Manipulations with Force Feedback in Complex Interaction Scenarios*

BSc & MSc in Mechanical Engineering @ Universidad de Navarra, Tecnun San Sebastian, 2002.09 — 2008.02

Erasmus and Leonardo scholarships · Exchange student at *Technische Universität München*, TUM (Munich)

Skills

Machine Learning, CV & NLP (expert): Pytorch & Lightning, Tensorflow & Keras, Scikit-Learn, Numpy, Pandas, SciPy, Matplotlib · OpenCV, Open3D, Halcon, PCL · NLTK, spaCy, HuggingFace Transformers & Diffusers, LangChain & LangGraph

Software Development (advanced): Python, C/C++, SQL, Bash · FastAPI, Flask, Streamlit, Gradio, HTML & CSS · Object Oriented Programming, Design Patterns, Data Structures, System Design

Cloud, MLOps & DevOps (working knowledge): AWS, Azure, PySpark, Elasticsearch · Docker & Kubernetes, Git & CI/CD, MLflow, DVC, Pytest & Linting Tools

Languages (proficient; certificates available): English (advanced - C1), German (advanced - C1), Spanish (native), Basque (native)

Honors and Awards

Publication Awards: Best Video Award Finalist (IEEE ICRA 2014) · Best Paper Award (VAMR 2013) · Short Paper Honorable Mention (IEEE VR 2012)

Academic Awards: Second Best Diploma (MSc) Thesis Prize 2008 (Association of Engineers of Gipuzkoa) · High School Excellence Award for University Access 2002 (Basque Government)